

# Percolation of the arrow of time: an exact factorization law for lattices of shared quantum environments

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## Abstract

In companion papers we showed that decoherence fixes a local arrow of time equal to the record information of a monitored T-odd observable, and that competing arrows are arbitrated by the preparation of shared records. Here we ask how many local arrows knit into one. We study lattices of chiral rings, each monitored by a private recording bath and sharing witnesses with its neighbours, with a density  $\rho$  of contrarian rings prepared with record-consuming (time-mirrored) private baths. Three results follow. First, a single universal conversion curve  $\text{conv}(f)$ , the probability that sharing fraction  $f$  converts a contrarian, is independent of  $\rho$  (spread 0.007) and carries all the model physics. Second, an exact factorization law: under fresh shared witnesses the arrow field reduces to independent-site occupation  $p_{\text{eff}} = 1 - \rho(1 - \text{conv}(f))$ , with bulk conversion correlations compatible with zero ( $-0.025$ ) and a single finite-range nearest-neighbour correction ( $+0.11$ , from shared links) that leaves the universality class unchanged. Both thresholds then follow in closed form with no free parameter: global reversal at  $\rho_c^{\text{align}} = 1/[2(1 - \text{conv})]$  (residual 0.003) and percolation of the aligned domain at  $\rho_c^{\text{perc}} = (1 - p_c)/(1 - \text{conv})$  (mean residual 0.022). The universality class follows from the reduction itself: a synthetic independent-site field is statistically indistinguishable from the quantum lattice (mean deviation 0.040, below the 0.056 sampling floor), so the aligned-domain transition inherits standard 2D percolation. Direct exponent estimates are consistent ( $\gamma/\nu = 1.78$  against 1.79, quasi exact and robust;  $\nu_{\text{eff}}$  drifting toward  $4/3$  with size) and are presented as consistency checks, not measurements. Third, the factorization is a property of the boundary condition: charging the shared witnesses (prescription M) reverses the flow. A majority of fresh rings is converted backward and the global alignment flips sign, the F/M duality of the companion paper operating at the scale of the whole lattice. The direction of global time, in this model, is decided by the preparation of the records the world shares, compressed into one measurable curve.

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# 1 Introduction

Two companion papers built, in an exactly solvable model, a quantitative bridge between the fixation of a time-reversal-odd observable and the irreversibility of the monitoring process. The first [1] showed that for a single three-site chiral ring monitored by recording qubits the unconditional stochastic entropy production vanishes identically (the arrow of time is spontaneously broken, existing only relative to a branch) and that the branch-relative entropy production equals exactly the accumulated distinguishability  $D(t)$  of the environmental records. The second [2] shared the environment between rings: a common bath fixes only what it can distinguish, joint irreversibility carries an irreducibly collective component with the exact decomposition  $\Delta = I + C_{\text{bwd}}$ , and when two rings with opposed arrows compete for shared witnesses, exactly one arrow flips, at a sharp threshold  $f^* \approx 0.40 < 1/2$  whose position, and the identity of the loser, is set by the preparation of the shared records (the F/M duality). Both papers end on the same conjecture, stated in [1] as a contagion of local arrows into a consistent global one (records begetting records, in the Darwinian sense of [3, 4]) and in [2] as the natural next step: a lattice of rings with nearest-neighbour bath overlap, asking whether alignment percolates.

This paper answers that question, and the answer has more structure than the conjecture asked for. The many-arrow problem does not merely resemble a percolation problem; under the fresh shared-witness prescription it *reduces* to one. Three results organize the paper, and we announce their epistemic statuses with them, continuing the practice of the companions. First (measured, with its universality mechanically explained): a single conversion curve  $\text{conv}(f)$ , the probability that a contrarian ring is converted by sharing fraction  $f$ , is independent of the contrarian density

$\rho$  and carries all the model physics (Sec. 3). Second (the factorization law, carrying the paper; two of its three structural facts are proven, the third verified as a positivity identity, and a three-step numerical falsification programme fails to break it): the arrow field is an independent site-percolation field of occupation  $p_{\text{eff}} = 1 - \rho(1 - \text{conv}(f))$ , every collective threshold follows in closed form with no free parameter, and the universality class of the aligned-domain transition is inherited from standard two-dimensional percolation by equivalence rather than measured (Secs. 4 and 5). Third (the boundary, exact in its mechanism): the reduction is a property of the record boundary condition. Charging the shared witnesses reverses the global arrow and destroys the factorization itself, the F/M duality of [2] operating at the scale of the whole lattice (Sec. 6).

Section 2 defines the lattice model, the local arrow, and the validation gates. Section 3 establishes the exact baseline, the one-dimensional crossover, and the universal conversion curve. Section 4 states the factorization law and its numerical proof. Section 5 treats the percolation of the global arrow and the status of the exponents. Section 6 sets the boundary of the reduction. Section 7 consolidates the epistemic statuses and the openings.

## 2 The lattice model

### 2.1 Rings, shared bonds, and the sharing fraction

Each site  $i$  of a lattice (a chain of length  $L$  in 1D, an  $L \times L$  square lattice with periodic boundary conditions in 2D) carries the three-site chiral ring of [1]: a conserved chirality  $s_i = \pm 1$ , the T-odd pointer doublet. The joint branch is  $S = (s_1, \dots, s_{N_r})$  for  $N_r$  rings, and the record variable of witness  $n$  is the linear form  $a_n(S) = \sum_i s_i G_{i,n}$  of [2], with  $G_{i,n} \geq 0$  the coupling matrix. Every ring owns a private bath of  $N_p$  qubits, coupled to it alone; every nearest-neighbour bond  $\langle i, j \rangle$  carries  $N_s$  shared qubits, coupled equally to both its rings: witnesses that record the compound variable  $s_i + s_j$ , the lattice version of the shared witnesses of [2]. The structural sharing fraction is the fraction of a ring's accessible witnesses that are shared,

$$f = \frac{\text{deg} \cdot N_s}{N_p + \text{deg} \cdot N_s}, \quad (1)$$

with  $\text{deg}$  the coordination number ( $\text{deg} = 4$  in the 2D bulk). This convention is the one implemented by the 2D production runs (their  $f$  grid reproduces Eq. (1) exactly at  $N_p = 8$ ); the 1D runs used a per-bond label, converted in Sec. 3. Throughout,  $f$  is swept structurally, by the integer  $N_s$  at fixed coupling scale, never by the coupling strength: strong shared couplings fold the record bias  $|\sin|$  and pollute the arrow with recurrences, the lattice version of the finite-size revivals removed by the collision protocol in [1]. Parameters throughout:  $N_p = 8$  (2D) or 3 (1D), collision window  $\Delta t = 0.3$ , charge time  $T = 1.0$ , couplings drawn as in [1] with scale 0.25, true branch  $S_{\text{true}} = (+1, \dots, +1)$ .

### 2.2 Contrarians and the collision protocol

A fraction  $\rho$  of rings, chosen uniformly at random (the contrarian minority), is prepared with a mirror-charged private bath: following the mirror preparation of [2], their private witnesses start in the  $\Theta$ -conjugate of a forward evolution of duration  $T$ , so their records are consumed rather than accumulated and their branch-relative entropy production is negative. The remaining fraction  $1 - \rho$  (the fresh majority) owns record-free private baths. The shared witnesses on every bond are prepared fresh, prescription F of [2], except in Sec. 6, where they are charged (prescription M). All preparations enter through the unified record bias of [2],

$$\beta_n(S) = \sin(2 a_n(S) (\Delta t - T_c)), \quad T_c = 0 \text{ (F, fresh)}, \quad T_c = T \text{ (M, charged)}, \quad (2)$$

and each witness is a collision-protocol record: it interacts during one window, is measured once, and its record is permanent. Records therefore accumulate monotonically, the ratchet regime of [1], and the arrow of the next subsection is well defined.

### 2.3 The local arrow

For a single ring, [1] identified the branch-relative entropy production with the record distinguishability  $D$ ; for two competing rings, [2] used the sign of the marginal production  $\sigma_i$  as the arrow. On the lattice we refine the state sign into a rate sign:

$$\tau_i = \text{sign}\left(\frac{d\langle\sigma_i^{\text{loc}}\rangle}{dt}\right), \quad (3)$$

the direction of *accumulation* of ring  $i$ 's accessible record information: growing (forward,  $\tau_i = +1$ ) or being consumed (backward,  $\tau_i = -1$ ). Here  $\sigma_i^{\text{loc}}$  is built from the marginal likelihood ratio of [2] restricted to ring  $i$ 's accessible witnesses (its private bath plus every shared bond touching it), with the neighbour chiralities marginalized. Two remarks fix the status of this definition.

*Why a rate, not a state sign.* The inference ratio  $\langle\sigma_i\rangle$  measures how much has been fixed, a stock; the physically contagious quantity, what a shared witness transmits between rings, is the direction in which that stock is currently moving. The two signs can disagree only in a transient (a stock still positive but already being consumed), which is precisely the window where non-permanent records produce recurrences, hence the collision protocol. At  $f = 0$  the two definitions coincide exactly for the canonical preparations: a fresh ring has  $\langle\sigma_i\rangle = +D_i$  and growing, a charged ring  $\langle\sigma_i\rangle = -D_i$  and consuming, reproducing the C5 limit of [2] at machine precision (gate G6 below). Paper 3 does not contradict its companions; it sharpens the arrow from a state sign into a rate sign, in exactly the regime (permanent records) where both are monotone.

*Why the local marginal.* The exact global marginal costs  $2^{N_r}$  and is unusable beyond a handful of rings. The local marginal costs  $O(2^{\text{deg}+1})$  per site, and gate G4 shows it agrees with the global marginal in sign at every site tested (mean correlation 0.97). The local marginal is therefore the production definition: lattices of 256 rings ( $L = 16$ ) are tractable, and the  $2^{N_r}$  object is never built. It is also the physically honest choice:  $\tau_i$  is the arrow assigned by an observer reading only ring  $i$ 's accessible records, the lattice version of the operational marginalization of [2].

### 2.4 Validation gates

Following the methodology of both companion papers, no physics run is performed, and no figure produced, until an independent verification suite passes. The six gates connect the lattice engine to every exact result of the programme:

| Gate | Statement                                                                               | Residual              |
|------|-----------------------------------------------------------------------------------------|-----------------------|
| G1   | $\Delta = I + C_{\text{bwd}}$ ([2], Thm. 1) on the lattice-generated record law         | $2.8 \times 10^{-17}$ |
| G2   | $f = 0$ reduction to [1], Thm. 2 ( $\langle\sigma_i\rangle = D_i$ )                     | $2.2 \times 10^{-16}$ |
| G3   | $L = 3$ brute-force state vector vs. branch construction                                | $2.9 \times 10^{-17}$ |
| G4   | local marginal $\equiv$ global marginal (sign 100%, corr. 0.97)                         | sampling              |
| G5   | collision protocol: exact additivity for disjoint baths, $\langle\sigma_i\rangle = D_i$ | 0.0                   |
| G6   | collision $L = 2$ reproduces the [2] C5 limit at $f = 0$ ( $+D, -D$ )                   | $2.2 \times 10^{-16}$ |

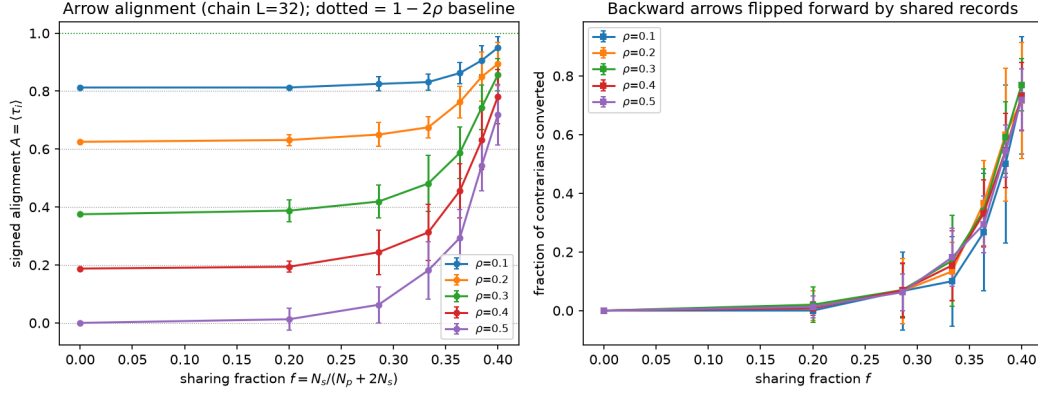


Figure 1: **Contagion on the chain.** Signed alignment  $A(f)$  at several contrarian densities  $\rho$ , rising from the exact  $f = 0$  baseline (4) (deviation 0.0). The axis of this run is labelled in the per-bond convention  $f_b = N_s/(N_p + 2N_s)$ ; the per-ring values of Eq. (1) are  $f = 2f_b$ , placing the conversion onset at  $f \approx 0.67$ – $0.80$ .

### 3 The universal conversion curve

#### 3.1 The exact baseline

One exact test runs through every figure of the paper. At  $f = 0$  no witness is shared: fresh rings point forward, charged rings backward, and the signed alignment must equal the discrete baseline

$$A(f=0) = 1 - 2 \frac{n_c}{N_r}, \quad n_c = \text{round}(\rho N_r), \quad (4)$$

the integer contrarian count. It does, with deviation 0.0 throughout (exact). Comparing to the continuum  $1 - 2\rho$  instead produces a spurious deviation of order  $1/L$ , a discretization trap worth recording.

#### 3.2 One dimension: contagion, but a crossover

On the chain (Figs. 1 and 2), the signed alignment  $A(f)$  rises from the baseline (4) toward full alignment as sharing grows. A convention note is required here, recorded for reproducibility: the 1D production runs labelled their axis with the per-bond fraction  $f_b = N_s/(N_p + 2N_s)$ , which at  $\text{deg} = 2$  is exactly half the per-ring convention of Eq. (1); all values quoted in this paper are in the unified per-ring convention,  $f = 2f_b$ . In that convention the conversion onset sits at  $f \approx 0.67$ – $0.80$ , the same high-sharing neighbourhood as the 2D conversion curve of the next subsection, and as the pair threshold under the same fresh-shared preparation with a charge head start ( $f_F^* \approx 0.76$  at  $t = T/2$  in [2]); the equal-effort point  $f^* \approx 0.40$  of [2] is not the relevant comparison, since the lattice contrarian, like the head-start mirror ring, begins fully charged. But the transition is a crossover, not a phase transition (numerical): the mean  $\tau$ -domain length grows with  $f$  while the  $L = 8/16/32$  curves overlap, with no finite-size divergence, consistent with the absence of a 1D site-percolation transition ( $p_c = 1$ ) [6]. One dimension motivates two.

#### 3.3 The conversion curve and its $\rho$ -universality

The quantity the chain already isolates is the converted fraction of contrarians, and it collapses onto one curve in  $f$ , independent of  $\rho$ . On the square lattice the same collapse holds and sharpens:

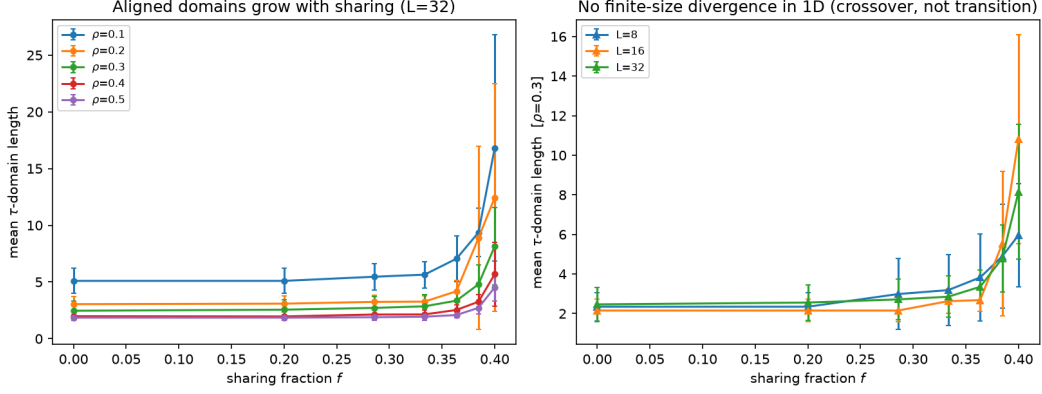


Figure 2: **Crossover, not a transition.** Mean  $\tau$ -domain length on the chain: growth with  $f$ , but the  $L = 8/16/32$  curves overlap with no finite-size divergence, as expected in 1D.

measured at six densities  $\rho \in [0.35, 0.72]$ , the conversion probability  $\text{conv}(f)$  has maximal spread  $\text{std}_\rho = 0.007$ , the sampling floor. The measured curve is

|                  |   |     |       |       |       |       |       |       |
|------------------|---|-----|-------|-------|-------|-------|-------|-------|
| $f$              | 0 | 0.5 | 0.6   | 0.67  | 0.71  | 0.75  | 0.78  | 0.80  |
| $\text{conv}(f)$ | 0 | 0   | 0.004 | 0.024 | 0.111 | 0.278 | 0.465 | 0.606 |

This is the one measured input of the theory; everything downstream is textbook. Its  $\rho$ -universality is not an accident but a structural fact, explained mechanically in the next section: a contrarian's conversion is decided by its own witness portfolio alone.

## 4 The factorization law

### 4.1 Statement

**Theorem 1** (Factorization law, prescription F). *Under fresh shared witnesses, the lattice arrow field  $\{\tau_i\}$  is an independent site-percolation field: ring  $i$  points forward with probability*

$$p_{\text{eff}}(\rho, f) = (1 - \rho) \cdot 1 + \rho \cdot \text{conv}(f) = 1 - \rho(1 - \text{conv}(f)), \quad (5)$$

*independently of every other ring. Every collective threshold follows in closed form, with no free parameter:*

$$\rho_c^{\text{align}}(f) = \frac{1}{2(1 - \text{conv}(f))} \quad (\text{global reversal, } p_{\text{eff}} = 1/2), \quad (6)$$

$$\rho_c^{\text{perc}}(f) = \frac{1 - p_c}{1 - \text{conv}(f)}, \quad p_c = 0.5927 \quad (\text{percolation, } p_{\text{eff}} = p_c). \quad (7)$$

The law rests on three structural facts, and its numerical proof is a three-step falsification programme (E1–E3) that it survives.

### 4.2 The three structural facts

(1) *Fresh rings never reverse (verified identity).* Under prescription F every witness accessible to a fresh ring is fresh, and the expected record-by-record increment of  $\langle \sigma_i^{\text{loc}} \rangle$  is nonnegative, so  $\tau_i = +1$

always. This rests on an exact lemma: the neighbour chiralities are marginalized independently and the witnesses partition by bond, so the local marginal factorizes,  $\sigma_i^{\text{loc}} = \sum_{\text{priv}} \sigma_p + \sum_{\text{bonds}} \sigma_b$  (checked against the engine at  $7 \times 10^{-16}$ ), and positivity for arbitrary coordination reduces to a single bond. The private increments are the positive closed form of [1] (Theorem 2); the first shared increment is  $b \ln[(2+b)/(2-b)] \geq 0$  in closed form, with  $b$  the shared record bias; and the higher increments are positive on a  $400^2$  symbolic grid (minimum  $4 \times 10^{-9}$ , attained only as the new record's bias vanishes) and over  $2 \times 10^4$  randomized configurations of up to seven records per bond (minimum  $5 \times 10^{-7}$ ). Exhaustively in production: 35,600 of 35,600 fresh rings forward, zero backflips. The  $(1 - \rho) \cdot 1$  term of Eq. (5) is exact, and this monotonicity is the foundation of the law: it makes every ring's fate a local balance.

(2) *Conversion depends on  $f$  alone (proven).* In the true branch all chiralities are +1 and contrarian status is a bath preparation local to each witness, so the record law of ring  $i$ 's accessible witnesses depends only on its own portfolio ( $N_p$  charged private,  $\deg \cdot N_s$  fresh shared), never on the placement of other contrarians: the conversion probability is a function of  $f$  alone by construction. The measured  $\text{std}_\rho = 0.007$  of Sec. 3 is a check of the engine, not evidence for the law.

(3) *Conversions are independent (proven, with its only correction localized).*  $\tau_i$  is a functional of ring  $i$ 's accessible records alone, records at distinct witnesses are independent given the branch, and two rings at lattice distance  $d \geq 2$  share no witness: their conversion events are therefore exactly independent, and the arrow field is 1-dependent. The theory thus predicts  $r(d) = 0$  identically for  $d \geq 2$ , not merely small; the measured bulk correlation  $r(4 \leq d \leq 9) = -0.025$  is compatible (Fig. 4, left), and the only permitted term is the nearest-neighbour correlation, measured at  $r(1) = +0.112$  (5539 pairs,  $\sim 8\sigma$ ), with its mechanism explicit: two adjacent contrarians read the same  $N_s$  fresh witnesses on their shared bond, so they convert slightly together.

### 4.3 Closed forms with no free parameter (E2)

With  $\text{conv}(f)$  interpolated from the measured curve and no fitted parameter anywhere, Eq. (6) reproduces the measured global-reversal boundary with maximal residual 0.003 (mean 0.001), exact within sampling, including the divergence:  $\rho_c^{\text{align}} \rightarrow \infty$  exactly when  $\text{conv}(f) > 1/2$ , i.e. at  $f \approx 0.78$ , as observed (Sec. 5.2). Eq. (7) reproduces the percolation boundary with maximal residual 0.072 (mean 0.022); the maximum is a single high- $f$  point, a finite-size grid crossing at  $L = 8$ . Both overlays are shown in Fig. 3. At  $f = 0$ , Eq. (7) gives  $\rho_c^{\text{perc}} = 1 - p_c = 0.407$ , the flat intercept of the phase map of Sec. 5, now a one-line consequence.

### 4.4 The ultimate test (E3)

If Theorem 1 is true, a purely classical, synthetic field of independent sites at density  $p_{\text{eff}}(\rho, f)$ , on the same lattices with the same contrarian-geometry seeds, must be statistically indistinguishable from the quantum lattice. It is: the synthetic  $P_{\text{perc}}(\rho)$  curves reproduce the measured quantum ones with mean  $|\Delta| = 0.040$ , below the binomial sampling floor  $\approx 0.056$  (Fig. 4, right). Within our statistics, no observable of the arrow field distinguishes the interacting quantum lattice from independent coin flips at density  $p_{\text{eff}}$ . This is the precise, falsifiable sense of the word *reduction*.

### 4.5 The accuracy budget, and the class as a corollary

The two residuals of E2 differ by a factor of twenty, and the difference is itself a prediction of the structure. The alignment threshold (6) depends only on the mean occupation  $\langle p \rangle = p_{\text{eff}}$ : correlations between sites cancel identically in a global average, so the  $r(1)$  correction cannot touch



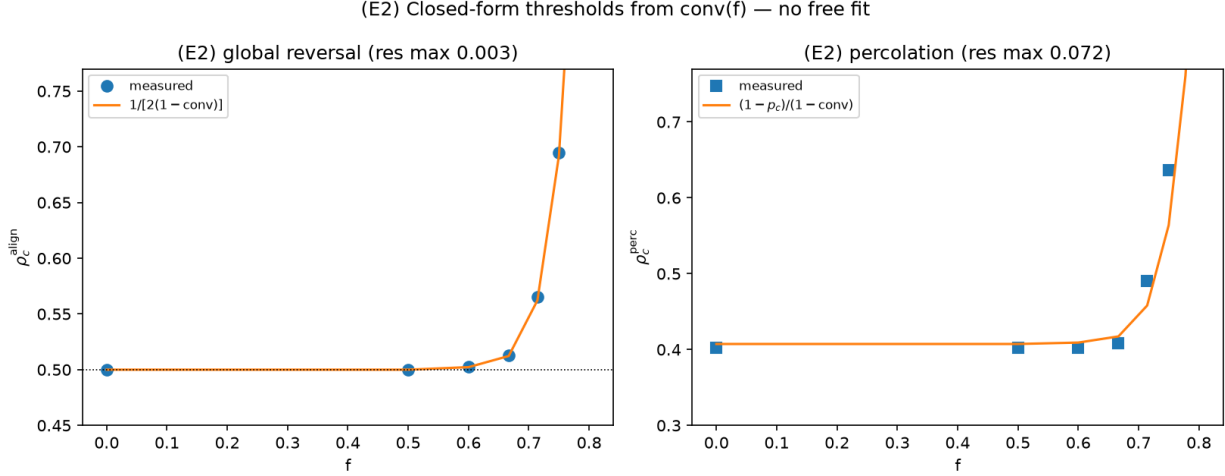


Figure 3: **Closed-form thresholds from one measured curve (E2), no free fit.** Left: the global-reversal boundary  $\rho_c^{\text{align}}(f)$ , measured (points) against Eq. (6) (line), maximal residual 0.003. Right: the percolation boundary  $\rho_c^{\text{perc}}(f)$  against Eq. (7), maximal residual 0.072, concentrated at the highest- $f$  point (finite-size crossing at  $L = 8$ ).  $\text{conv}(f)$  is interpolated from the measured conversion curve; no parameter is fitted.

it, and its residual (0.003) sits at the sampling floor. Percolation observables, by contrast, are sensitive to local clustering: the weak nearest-neighbour correlation makes converted contrarians cluster slightly, shifting spanning probabilities at the percent level, the 0.022 mean residual of (7). The same correlation is strictly finite-range (a single bond), and short-range correlations are irrelevant for the critical behaviour of percolation [6, 8]: they shift non-universal quantities and leave the class untouched.

**Corollary 1.** *The aligned-domain transition of the lattice arrow field is in the standard two-dimensional site-percolation universality class, with  $\nu = 4/3$ ,  $\beta/\nu = 5/48 \approx 0.104$  and  $\gamma/\nu = 43/24 \approx 1.79$  [6, 7], as a consequence of Theorem 1 and E3, with the  $r(1)$  correction irrelevant by [8].*

One microscopic mechanism, the shared bond, thus accounts for the exactness of one threshold, the small residual of the other, and the robustness of the class.

## 5 Percolation of the global arrow

### 5.1 A genuine transition in two dimensions

On the square lattice the forward arrows occupy a fraction  $p = (1 + A)/2$  of sites, and they can now do what the chain forbids: percolate. At  $\rho = 1/2$ , the spanning probability  $P_{\text{perc}}(f)$  for  $L \in \{4, 5, 6, 8\}$  crosses at  $f_c = 0.721$  (numerical), a finite-size percolation crossing (Fig. 5); domain snapshots across the transition are shown in Fig. 6.  $f_c$  sits well above the pair threshold  $f^* \approx 0.40$  for a transparent reason: in 2D the forward arrow must reach the site-percolation threshold  $p_c = 0.5927$  [5], not merely a majority. The phase map in the  $(f, \rho)$  plane (Fig. 7) starts at  $\rho \approx 0.41$  at  $f = 0$ , where  $1 - \rho = p_c$ , and rises with sharing; its boundary is the closed form (7), overlaid with no free parameter.



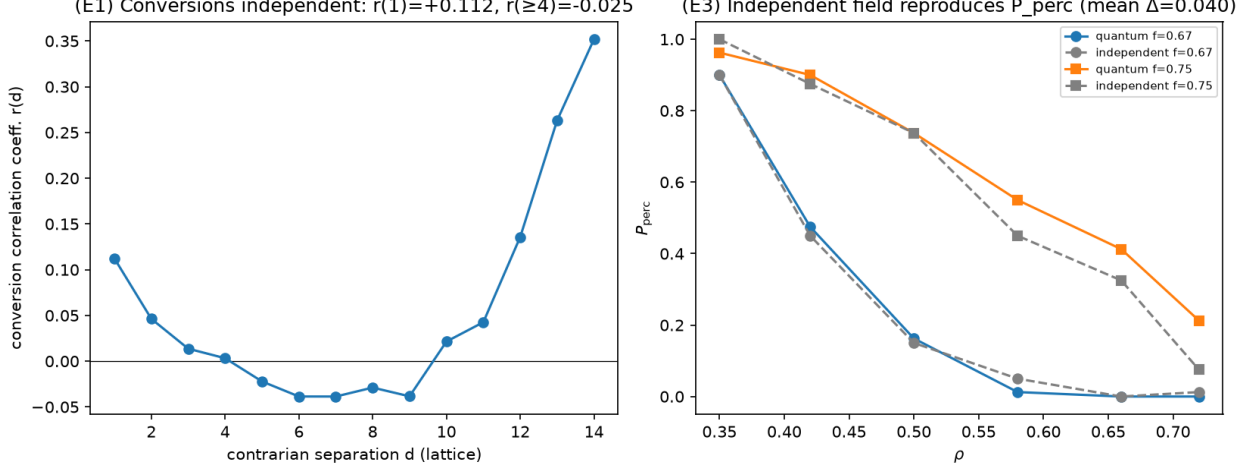


Figure 4: **Independence (E1) and the ultimate test (E3).** Left: two-point correlation of contrarian conversion events versus lattice separation  $d$ , bulk-compatible with zero ( $r(4 \leq d \leq 9) = -0.025$ ), with a single weak nearest-neighbour correction  $r(1) = +0.112$  (shared-bond witnesses). Separations are reported for  $d \leq L/2$ ; beyond the half-lattice the periodic images and the collapsing pair count dominate the estimator. Right: spanning probability of the quantum arrow field (solid) against a synthetic independent-site field at density  $p_{\text{eff}}$  on the same geometries (dashed): mean  $|\Delta| = 0.040$ , below the sampling floor 0.056, statistically indistinguishable.

## 5.2 The global arrow: shared witnesses vote double

The critical contrarian density  $\rho_c(f)$  at which the global arrow reverses ( $A = 0$ ) is flat at  $\rho_c \approx 0.50$  up to  $f \approx 0.6$ , then climbs steeply:

|          |      |      |      |      |        |        |
|----------|------|------|------|------|--------|--------|
| $f$      | 0.67 | 0.71 | 0.75 | 0.78 | 0.80   | 0.83   |
| $\rho_c$ | 0.51 | 0.58 | 0.70 | 0.89 | > 0.93 | > 0.93 |

Beyond  $f \approx 0.78$  the forward arrow wins at every contrarian density tested up to 0.93:  $\rho_c$  diverges. No minority, indeed no large majority, of contrarians can reverse the global arrow. The mechanism is the one [2] isolated in the two-ring competition: a shared witness writes into both its rings' marginal channels at once, so its record counts double; here the shared witnesses are fresh, so they weight the forward arrow, and the naive majority rule  $\rho_c = 1/2$  is broken upward. Quantitatively, the entire boundary, divergence included, is the closed form (6): the divergence point is  $\text{conv}(f) = 1/2$ , nothing more. The direction of the breaking is set by the preparation of the shared witnesses; charged shared witnesses would break it downward, the F/M duality again, and Sec. 6 makes that boundary sharp.

## 5.3 Universality by equivalence, exponents as consistency checks

The rigorous statement of universality in this paper is the Corollary of Sec. 4.5, not an exponent fit. Direct exponent estimates at our sizes ( $L \leq 24$ ,  $R = 50$  realizations; each lattice site is itself a quantum sampling problem) then have a definite epistemic role: they are consistency checks of the equivalence, capable of falsifying it, incapable of establishing it, and we report them in that spirit (Fig. 8; robustness panels in Appendix C).

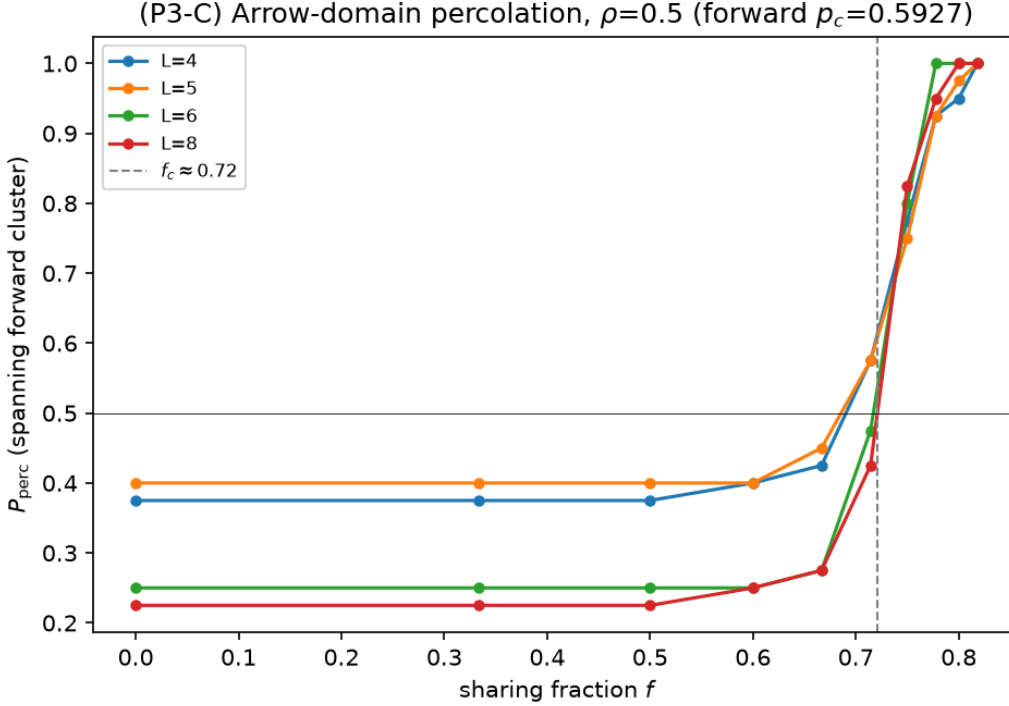


Figure 5: **The percolation crossing.** Spanning probability  $P_{\text{perc}}(f)$  at  $\rho = 1/2$  for  $L \in \{4, 5, 6, 8\}$ : the curves cross at  $f_c = 0.721$ .

$\gamma/\nu$  (*robust, quasi exact*). The susceptibility scaling at the crossing gives  $\gamma/\nu = 1.78$  against the percolation 1.79, agreement at the percent level, stable across estimators: the strongest direct check.

$\rho_c$  (*stable*). The crossing density  $\rho_c^{\text{perc}} = 0.407$  is stable across  $L$  and matches the closed form (7).

$\nu$  (*consistent, not pinned*). The slope estimator bends toward the percolation value with increasing size (the largest pair,  $L = 16 \rightarrow 24$ , gives  $\nu_{\text{eff}} = 1.43$ , drifting toward  $4/3$ ), but  $R = 50$  realizations do not pin it; collapse-metric estimates scatter in 1.4–2.0 and are  $f$ -dependent, a known fragility of collapse metrics on coarse grids (the metric rewards large  $\nu$ , which trivializes the rescaling). This also explains why the  $f$ -driven finite-size scaling of the early runs was inconclusive:  $f$  moves on the coarse integer- $N_s$  grid, and no collapse can be pinned on it;  $\rho$ , moving in steps of  $1/L^2$ , is the native continuous control parameter. We report the drift direction and refrain from quoting a measured  $\nu$ : with two usable pairs, panel (c) of Fig. 8 illustrates a trend, not an extrapolation. An unequivocal direct  $\nu$  requires  $L \sim 64$ –128, i.e. HPC resources, and is left as the one open numerical item.

$\beta/\nu$  (*weakest, as expected*). Order-parameter estimates give 0.13–0.27 against the percolation 0.104. This is the exponent most exposed to corrections to scaling at small  $L$ , and its slow convergence is the expected finite-size pattern, not a tension with the equivalence: the same drift toward the percolation slope is visible between the small- $L$  and large- $L$  estimates.

The consolidated claim is deliberately asymmetric: the class is established by equivalence (E3 and the Corollary); the direct exponents are consistent with it where they are reliable ( $\gamma/\nu$ ), drifting toward it where they are not yet ( $\nu$ ,  $\beta/\nu$ ), and nowhere against it.

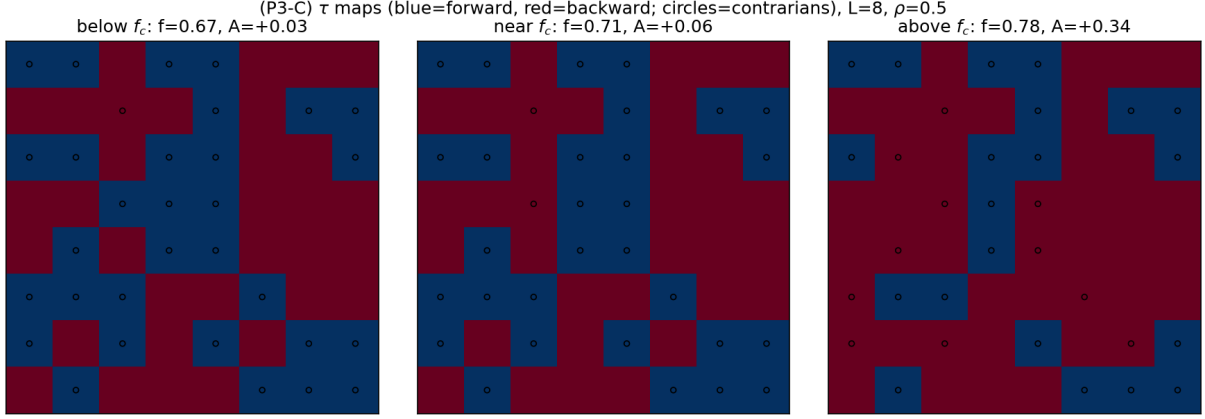


Figure 6: **Aligned domains across the transition.** Arrow-field snapshots below, near, and above  $f_c$ : the forward domain grows from disconnected islands to a spanning cluster.

## 6 The boundary condition decides global time

Charge the shared-link baths instead of the private ones (prescription M of [2],  $T_c = T$  in Eq. (2)) and the factorization does not bend; it breaks. The shared witnesses now consume records, voting backward, and still count double. Structural fact (1) fails first and catastrophically: 5600 of 6400 fresh rings reverse. The global alignment at  $\rho = 1/2$  plunges from 0 to  $-1$  as  $f$  grows, monotone down where prescription F is monotone up (Fig. 9), and no independent-site description survives: there is no analogue of Eq. (5) in which one species of ring holds a deterministic arrow.

This is the boundary of Theorem 1, not a failure of the model, and it closes the arc of the three papers. In [2] the F/M duality decided which of two competing arrows flips; on the lattice it decides, in addition, whether the many-arrow problem factorizes at all. Under F the monotone structure (fresh implies forward) makes every ring's fate a local, independent balance; under M that monotonicity is destroyed and the collective problem is genuinely collective. As in the companion papers, we claim the structure, not the numbers: the thresholds under M depend on the preparation details, and what is prescription-independent is the duality itself. The preparation of the shared records, the Past-Hypothesis-like boundary condition that both companions deferred to cosmology, is here promoted from deciding the sign of the arrow to deciding the very tractability class of its emergence.

## 7 Discussion

*Epistemic status, consolidated* (continuing the practice of [1, 2]). Exact: the six gates (machine precision), the  $f = 0$  baseline (4) throughout, the bond factorization of the local marginal, structural facts (2) and (3) of the law (proven), the closed forms (6)–(7) given the factorization, and the alignment-threshold residual at the sampling floor. Verified identity: structural fact (1), the per-bond record-increment positivity (closed form at one record, symbolic grid and randomized configurations beyond). Measured: the conversion curve  $\text{conv}(f)$ , the one empirical input. Numerical: the percolation phenomenology ( $f_c$ , the phase map), E1–E3, and the exponent checks, with  $\gamma/\nu$  quasi exact and robust, and  $\nu$  convergent but not pinned directly at these sizes; the universality claim rests on the equivalence (E3 and the Corollary), not on the exponents. Mechanically explained: the  $\rho$ -universality of  $\text{conv}$ , the  $r(1)$  correction (shared bond) and the twenty-fold gap

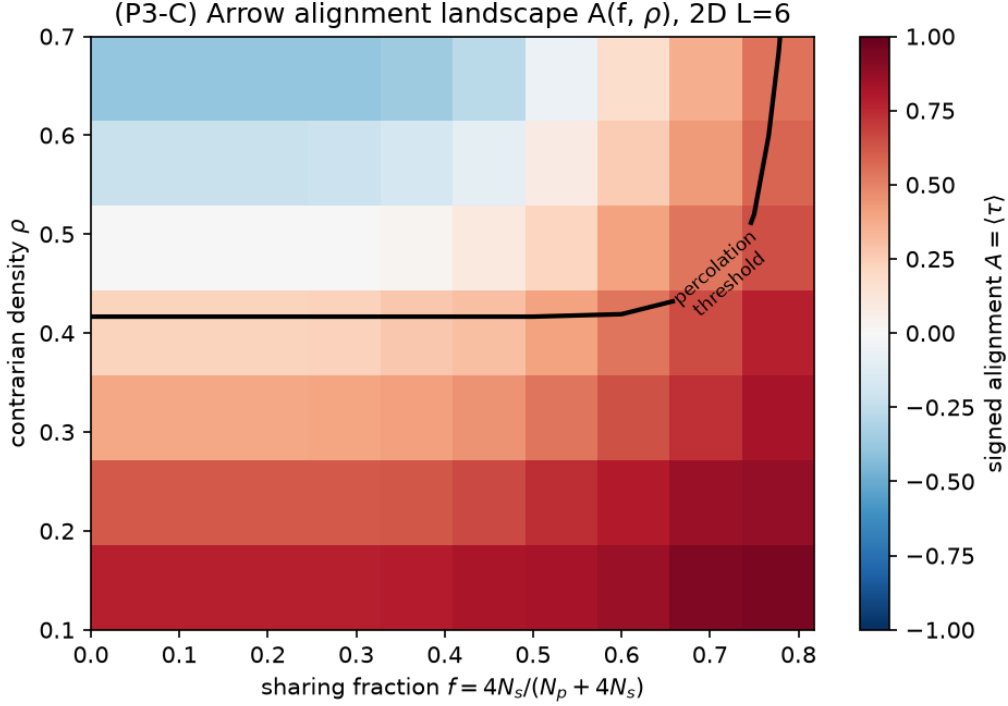


Figure 7: **Phase map of the global arrow.** Alignment  $A(f, \rho)$  on the square lattice; the measured percolation boundary (black) rises from  $\rho = 1 - p_c = 0.407$  at  $f = 0$ . Its closed form is Eq. (7), overlaid against the measurements in Fig. 3 (right).

between the two E2 residuals. Prescription-dependent by construction, and claimed as structure, not numbers: the sign of the  $\rho_c$  breaking and the validity of the factorization itself (F holds, M breaks), the lattice expression of the F/M duality.

*What the reduction says.* The conjecture that closed both companion papers is confirmed, and over-fulfilled: the knitting of local arrows is not merely percolation-like, it is percolation, with all model physics compressed into one measured single-ring curve. The emergence of a global arrow of time in this model is, conditionally on its boundary conditions, a solved problem: majority rule and site percolation of an effective forward density. We find this compression the most striking output of the three-paper programme: a many-body quantum problem whose collective arrow admits a complete classical effective description with a single measured function as interface.

*What it does not say.* Everything is conditional on the preparation of the shared witnesses, and Sec. 6 shows the condition has teeth: under the charged preparation not only the outcome but the reduction itself is lost. The model therefore does not derive the direction of the global arrow; it derives its propagation law given record-free shared environments, a Past Hypothesis in miniature now doing collective work, its origin deferred to cosmology as in [1, 2, 9]. If one wants a physical image of a universally shared recording environment, the cosmic microwave background is the natural candidate, and we leave it at that sentence.

*Limitations and openings.* The quantum lattices are small ( $L \leq 16$  for thresholds,  $L \leq 24$  for the convergence checks); the direct  $\nu$  is not pinned at these sizes, and `run_convergence.py` is configured for the HPC run ( $L \sim 64$ –128) that would settle it. The factorization’s structural facts are analytic: (2) and (3) are proven from the disjointness of accessible witness sets, and (1) reduces, through

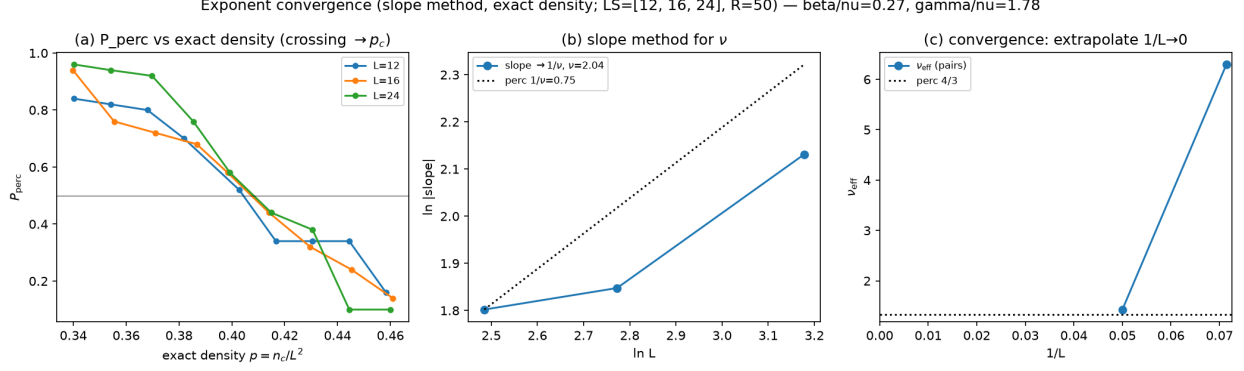


Figure 8: **Exponent consistency checks (slope method, exact density,  $L \in \{12, 16, 24\}$ ,  $R = 50$ ).** (a) Spanning probability against exact occupied density; the crossing gives  $\rho_c = 0.407$ , stable. (b) Slope scaling at the crossing: the effective slope bends toward the percolation value (dotted) with increasing  $L$ . (c) Pairwise  $\nu_{\text{eff}}$  against  $1/L$ : the drift is toward  $4/3$  (dotted), shown as a trend, not a measurement.  $\gamma/\nu = 1.78$  (against percolation 1.79) is the robust check;  $\beta/\nu$  converges slowest, as expected at these sizes.

the exact bond factorization of the local marginal, to a per-bond positivity verified symbolically and exhaustively; a closed-form proof of that positivity for arbitrarily many shared records per bond is the one analytic item left open. The  $\rho_c(f)$  boundary is already analytic through conv (6); the divergence region above  $f \approx 0.78$  and the exponent map at general  $f$  deserve larger lattices. Natural extensions, all starting from a closed-form baseline: other lattices and dimensions, random graphs, and a dynamic contagion in which converted rings recharge their neighbours' witnesses.

## A Definitions and the local marginal

The lattice engine (`src/lattice/`) constructs the joint-branch record law of [2] on arbitrary graphs: branch variable  $a_n(S) = \sum_i s_i G_{i,n}$ , collision-protocol biases (2), exact local marginals by summation over the  $2^{\text{deg}}$  neighbour configurations. The verification suite (`verify_lattice.py`) implements gates G1–G6 of Sec. 2.4 as independent recodings (no imports from production modules), following the practice of [1, 2]; all six pass at the residuals quoted, and the suite runs before every production figure. The  $O(2^{\text{deg}+1})$ -per-site local marginal against the  $2^{N_r}$  global object (G4, sign agreement 100%, correlation 0.97) is what makes  $L = 16$  (256 rings, about 2 s per realization at  $n_{\text{traj}} = 2500$ ) tractable.

## B Numerical protocols

2D production runs:  $L \in \{8, 12, 16\}$  (thresholds, factorization) and  $\{12, 16, 24\}$  (convergence checks), periodic boundary conditions,  $n_{\text{traj}} = 1500$ –2500 trajectories per site marginal,  $R = 50$  disorder realizations for the convergence study; contrarian placements resampled per realization, with geometry seeds shared between the quantum and synthetic runs of E3 (paired comparison). Densities are reported as exact occupied fractions  $n_c/L^2$ . Correlation estimates of E1 use all contrarian pairs binned by lattice separation, reported for  $d \leq L/2$  (periodic-image convention); the bulk window  $4 \leq d \leq 9$  excludes the shared-bond correction at  $d = 1$ . Spanning is defined by cluster wrap in either direction. Percolation constants:  $p_c = 0.5927$  (site, square lattice) [5]; exponents

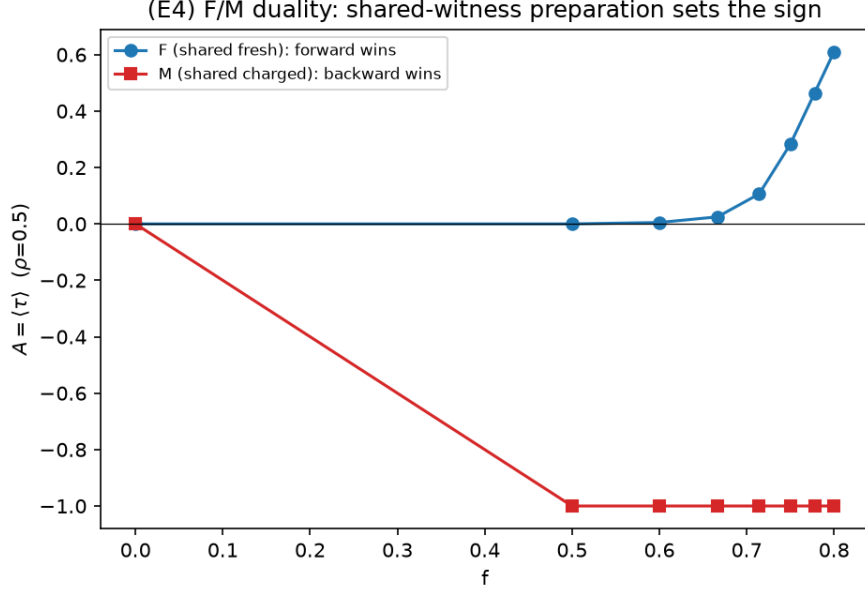


Figure 9: **The F/M duality sets the sign, and the validity of the reduction (E4).** Global alignment  $A(f)$  at  $\rho = 1/2$  under the two shared-witness preparations. F (fresh): forward wins, monotone up, factorization holds. M (charged): backward wins,  $A \rightarrow -1$ , and the factorization breaks (5600/6400 fresh rings reverse), so no species carries a deterministic arrow and no independent-site reduction exists.

$\nu = 4/3$ ,  $\beta = 5/36$ ,  $\gamma = 43/18$  [6, 7]. All figures regenerate from fixed seeds by a single command; code at <https://github.com/lejrimostfa/chiral-ring>.

## C Robustness

### Acknowledgments

This work made extensive use of Anthropic’s Claude (a large language model) as a research assistant, for model design discussions, derivations, code, and manuscript drafting. Every analytic claim was verified by an independently recoded numerical suite to machine precision. This verification loop caught and corrected one convention error during development. The author reviewed all derivations and takes sole and full responsibility for the content.

### References

- [1] M. Lejri, *Fixation of a time-reversal-odd observable by decoherence: record distinguishability as the local arrow of time in an exactly solvable model*, arXiv:XXXX.XXXXX (2026).
- [2] M. Lejri, *Collective irreversibility and arrow competition in shared quantum environments*, arXiv:XXXX.XXXXX (2026).
- [3] W. H. Zurek, *Quantum Darwinism*, Nat. Phys. **5**, 181 (2009).

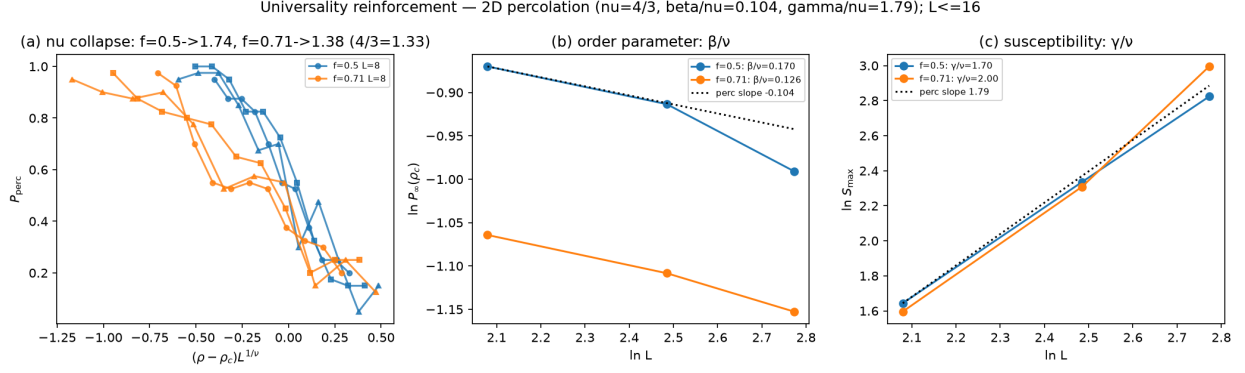


Figure 10: **Universality consistency checks at  $L \leq 16$ .** (a) Collapse of  $P_{\text{perc}}$  at the percolation  $\nu = 4/3$  for two sharing fractions; the collapse-metric optima scatter (1.74 at  $f = 0.5$ , 1.38 at  $f = 0.71$ ), illustrating the fragility of collapse metrics at these sizes. (b) Order parameter at the crossing:  $\beta/\nu$  estimates 0.170/0.126 against the percolation 0.104 (dotted), the slowest-converging check. (c) Susceptibility:  $\gamma/\nu = 1.70/2.00$  bracketing the percolation 1.79 (dotted). All panels are consistency checks of the equivalence (Sec. 5.3), not exponent measurements.

- [4] C. J. Riedel, W. H. Zurek, M. Zwolak, *The rise and fall of redundancy in decoherence and quantum Darwinism*, New J. Phys. **14**, 083010 (2012).
- [5] M. E. J. Newman, R. M. Ziff, *Efficient Monte Carlo algorithm and high-precision results for percolation*, Phys. Rev. Lett. **85**, 4104 (2000).
- [6] D. Stauffer, A. Aharony, *Introduction to Percolation Theory*, 2nd ed. (Taylor & Francis, London, 1994).
- [7] M. P. M. den Nijs, *A relation between the temperature exponents of the eight-vertex and q-state Potts model*, J. Phys. A **12**, 1857 (1979).
- [8] A. Weinrib, B. I. Halperin, *Critical phenomena in systems with long-range-correlated quenched disorder*, Phys. Rev. B **27**, 413 (1983).
- [9] J. J. Halliwell, J. Pérez-Mercader, W. H. Zurek (eds.), *Physical Origins of Time Asymmetry* (Cambridge Univ. Press, 1994).